# T3 final-EFT one-loop RGE term comparison: EFT_2_after_S1_S2

All T3 particles have been integrated out at this stage. The active theory is the SM plus the dimension-five Weinberg operator. Rows are the UV model configurations that produced the matched coefficient, while columns align the additive structures in its one-loop beta function. Each populated cell contains the complete term, including its coefficient and the factor $C_5$. We use $16\pi^2\,\mu\,dC_5/d\mu = \beta^{(1)}_{C_5}$.

## Notation key

| symbol | interaction / definition |
|---|---|
| $g_1$ | $D_\mu \supset i g_1 Y B_\mu$ |
| $g_2$ | $D_\mu \supset i g_2 T^A W_\mu^A$ |
| $g_3$ | $D_\mu \supset i g_3 t^A G_\mu^A$ |
| $Y_u$ | $-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$ |
| $Y_d$ | $-\bar{Q} Y_d H d + \text{h.c.}$ |
| $Y_e$ | $-\bar{L} Y_e H e + \text{h.c.}$ |
| $y_1$ | $L F S_1 + \text{h.c.}$ |
| $y_2$ | $L F S_2 + \text{h.c.}$ |
| $M_F$ | $-\bar{F} M_F F$ |
| $T$ | $\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$ | $\mathrm{Tr}(y_1^\dagger y_1)$ |
| $T_\nu^{(2)}$ | $\mathrm{Tr}(y_2^\dagger y_2)$ |
| $T_\nu$ | $T_\nu^{(1)} + T_\nu^{(2)}$ |
| $m_1^2$ | $m_1^2 S_1^\dagger S_1$ |
| $m_2^2$ | $m_2^2 S_2^\dagger S_2$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$ |
| $\lambda_{S_1}^{(1)}$ | $\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$ |
| $\lambda_{S_2}^{(1)}$ | $\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$ |
| $\lambda_{HS_1}^{(1)}$ | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$ |
| $\lambda_{HS_2}^{(1)}$ | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$ |
| $\lambda_{HS_1}^{(A)}$ | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$ |
| $\lambda_{HS_2}^{(A)}$ | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$ |
| $\lambda_5$ | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$ |
| $\lambda_{12}^{(1)}$ | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$ |
| $\lambda_{12}^{(A)}$ | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$ |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$ |
| $\lambda_{S_1}^{(A)}$ | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$ |
| $\lambda_{S_2}^{(A)}$ | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$ |
| $\alpha$ | $Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$ |
| $C_2(d)$ | $(d^2-1)/4$ |

| | |
|---|---|
| $T(d)$ | $d(d^2-1)/12$ |
| $C^{ij;AB}_{LLS_1S_2}$ | $L_iL_jS_1^AS_2^B$ |
| $C_{ijab}$ | $L_iL_j\phi_a\phi_b$ |
| $C_5^{ij}$ | $(L_i^TCL_j)HH$ |
| $R_W$ | $16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$ |

$\beta^{(1)}_{C_5}$

**Quantity:** $C_5$

$$16\pi^2\,\mu\frac{dC_5}{d\mu} = \beta^{(1)}_{C_5}$$

## Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $C_5 g_2^2$ |
| T2 | $C_5\lambda_1$ |
| T3 | $C_5 Y_e Y_e^\dagger$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-3C_5g_2^2$ | $2C_5\lambda_1$ | $-C_5Y_eY_e^\dagger$ |

**Quantity:** $C_5$

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $C_5 Y_d Y_d^\dagger$ |
| T2 | $C_5 Y_u Y_u^\dagger$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $6C_5 Y_d Y_d^\dagger$ | $6C_5 Y_u Y_u^\dagger$ |